

	GENERAL SCHEDULE						
	Tuesday	Wednesday	Thursday	Friday	Saturday		
8:00 - 8:30		Registration				8:00 - 8:30	
8:30 - 9:00		Opening				8:30 - 9:00	
9:00 - 9:30		Plenary Lecture Eugenio Roanes-Lozano	Plenary Lecture Ilias Kotsireas			9:00 - 9:30	
9:30 - 10:00		Parallel Sessions. Rooms I, II and III	Parallel Sessions. Rooms I, II and III	Parallel Sessions. Rooms I, II and III		9:30 - 10:00	
10:00 - 11:00						10:00 - 11:00	
11:00 - 11:30		Coffee Break				11:00 - 11:30	
11:30 - 13:00		Parallel Sessions. Rooms I, II and III			Parallel Sessions. Rooms I, II and III	11:30 - 13:00	
13:00 - 13:30						Closing	13:00 - 13:30
13:30 - 15:00		LUNCH					13:30 - 15:00
15:00 - 16:30		Parallel Sessions. Rooms I, II and III	Excursion to Granada visiting Alhambra with dinner	Parallel Sessions. Rooms I, II and III			15:00 - 16:30
16:30 - 17:00		Coffee Break		Coffee Break			16:30 - 17:00
17:00 - 18:30		Parallel Sessions. Rooms I, II and III		Parallel Sessions. Rooms I, II and III			17:00 - 18:00
18:30 - 19:00	Registration at loby of Hotel Málaga Palacio			ACA-WG Meeting			18:00 - 18:30
19:00 - 20:30		Excursion Málaga City Centre and Alcazaba with Dinner** in Restaurant "El Pimpi"					18:30 - 19:30
20:30 - 21:00					19:30 - 21:00		
21:00 - ?	Welcome reception and cocktail Town Hall*		Gala dinner at Hotel Málaga Palacio		21:00 - ?		

* Welcome reception time could be slightly different due to Mayor's availability

** Time for dinner will depend on the end of excursion

Plenary Lecture Room I	00. General Session on Computer Algebra Room III	01. Computer Algebra in Education Room II		02. Computer Algebra for Dynamical Systems and Celestial Mechanics Room III
Eugenio Roanes Lozano Some ideas about the research in computational mathematics: a perspective from the middle of the life and the academic life Wednesday 3, 9:00 - 10:00	A New Criterion for the Existence of Real Zeros of Polynomial Systems. J. S. Cheng Wednesday 3, 16:00 - 16:30	Computer Algebra - the engine of transition to activity-based approach in mathematics education E. Varbanova and E. Shoikova Friday 5, 9:30 - 10:00	In praise of rectangular systems D. Jeffrey Friday 5, 15:00 - 15:30	The Study of Isochronicity and Critical Period Bifurcations on Center Manifolds of 3-dim Polynomials Systems Using Computer Algebra M. Mencinger and B. Fercec Friday 5, 17:00 - 17:30
Ilias Kotsireas 10 years of sequence autocorrelation Plenary Lecture Thursday 4, 9:00 - 10:00	Planar arrangements and singular algebraic surfaces. J. García Escudero Wednesday 3, 17:00 - 17:30	Assessing Mathematical Content in a Technology Environment Discussion Panel A. Homero Flores Friday 5, 10:00 - 11:00	Omega - a free online Computer Algebra System Explorer M. Xue Friday 5, 15:30 - 16:00	On using computer algebra systems for analysis of rigid body dynamics L. Burlakova and V. Irtegov Friday 5, 17:30 - 18:00
Opening Ceremony Wednesday 3, 8:30 - 9:00	Computations of Gröbner-Shirshov Basis and Reduced Words for Affine Weyl Group $\tilde{A}_n$ using Mathematica. E. Yilmaz, Cenap Özel and Ugur Ustaoglu Wednesday 3, 17:30 - 18:00		Some maths problems for the average citizen E. Roanes-Lozano and J. Cabezas- Corchero Friday 5, 16:00 - 16:30	Formal Integral and Caustics in Henon- Heiles model T. Myllari and A. Myllari Saturday 6, 9:30 - 10:00
Closing Ceremony Saturday 6, 13:00 - 13:30		CAS: A Tool for Improving Autonomous Work A. García, F. García, Á. Martín, G. Rodríguez and A. de la Villa Friday 5, 11:30 - 12:00	Using Computer Algebra in Mathematics for Engineers T. Westermann Friday 5, 17:00 - 17:30	On necessary conditions of integrability of degenerated planar ODE systems in the parameter space A. Bruno and V. Edneral Saturday 6, 10:00 - 10:30
ACA-WG Meeting Friday 5, 18:00 - 19:30 (Room II)		Investigating Magic Squares in a Linear Algebra Course K. Schmidt Friday 5, 12:00 - 12:30	Global application of CAS tools for teaching in Computer Engineering degrees S. Cárdenas, I. Fortes, I. Pérez de Guzmán, S. Sánchez and A. Valverde Friday 5, 17:30 - 18:00	Tree structures in Poisson series processors J. F. Navarro Saturday 6, 10:30 - 11:00
		Polynomial Systems Solving with Nspire CAS (Part I, Part II) M. Beaudin, G. Picard and G. Savard Friday 5, 12:30 - 13:30		Orbital Reversibility of Dynamical Systems A. Algaba, I. Checa, C. García and E. Gamero Saturday 6, 11:30 - 12:00
				Normal Forms of Singular Plane Quartics T. Takahashi Saturday 6, 12:00 - 12:30
				Random walks on the three dimensional Young tree N. Vassiliev Saturday 6, 12:30 - 13:00

03. Algebraic and Algorithmic Aspects of Differential and Integral Operators Room I		04. Computer Algebra in Coding Theory and Cryptography Room I		05. Nonstandard Applications of Computer Algebra Room II	
Darboux theory of integrability in the sparse case G. Chèze Wednesday 3, 10:00 - 10:30	On Completely Integrable Pfaffian Systems with Normal Crossings S. Maddah Wednesday 3, 15:30 - 16:00	A characterization of cyclic codes whose minimum distance equals their maximum BCH bound J. J. Bernal, D. H. Bueno and J. J. Simon Thursday 4, 11:30 - 12:00	On Generalized Lee Weight Codes over Dihedral Groups E. Martínez-Moro, A. P. Nicolás and E. Suárez-Canedo Friday 5, 12:00 - 12:30	Simulating Car Traffic with Smart Signals using a CAS J. L. Galán, G. Aguilera, J. C. Campos and P. Rodríguez Wednesday 3, 15:00 - 15:30	A hybrid expert system for classic car recognition and originality evaluation E. Roanes-Lozano and J. Bonilla Thursday 4, 10:00 - 10:30
Isomorphisms and Serre's reduction of multidimensional linear systems T. Cluzeau and A. Quadrat Wednesday 3, 10:30 - 11:00	Linear Boundary Problems for Partial Differential Equations: Algebraic Setup and First Steps for Constant Coefficients N. Phisanbut and M. Rosenkranz Wednesday 3, 16:00 - 16:30	Gröbner Bases and Linear Codes over Prime Fields N. Dück and K. H. Zimmermann Thursday 4, 12:00 - 12:30	Decoding of codes for applications to steganography C. Munuera and W. Olaya León Friday 5, 12:30 - 13:00	Population-based anamorphosis maps for railway radial networks E. Roanes-Lozano, A. García-Álvarez, J. L. Galán-García and L. Mesa Wednesday 3, 15:30 - 16:00	Designing rotating schedules by using Gröbner bases R. Falcón, D. Canca and E. Barrena Thursday 4, 10:30 - 11:00
Qualitative Study of Polynomial Differential Systems D. Dali and A. Turqui Wednesday 3, 11:30 - 12:00	On the arithmetic of d'Alembertian functions C. Raab Wednesday 3, 17:00 - 17:30	A Class of Binary Sequences with Large Linear Complexity A. Fúster Sabater Thursday 4, 12:30 - 13:00	On LDPC codes corresponding to new families of regular expanding graphs of large girth M. Polak and V. Ustimenko Friday 5, 13:00 - 13:30	An algebraic approach to geometric proof using a Computer Algebra System M. Xue Wednesday 3, 16:00 - 16:30	Flexibility of Structures via Computer Algebra R. H. Lewis and E. Coutsias Thursday 4, 11:30 - 12:00
Periodic and Mean-Periodic Solutions of LODEs with Constant Coefficients I. Dimovski and M. Spiridonova Wednesday 3, 12:00 - 12:30	Applying Thomas decomposition and algebraic analysis to certain nonlinear PDE systems D. Robertz Wednesday 3, 17:30 - 18:00	Further Improvements on the Feng-Rao Bound for Dual Codes O. Geil and S. Martin Thursday 4, 13:00 - 13:30	Representation, constructions and minimum distance computation of binary nonlinear codes J. Pujol, M. Villanueva and F. Zeng Friday 5, 15:00 - 15:30	Designing Hamiltonian Cycles F. de Arriba, E. Corbacho and R. Vidal Wednesday 3, 17:00 - 17:30	Modeling reliability in propositions using computer algebra techniques A. Hernando Thursday 4, 12:00 - 12:30
Looking for invariant algebraic curves A. Ferragut and A. Gasull Wednesday 3, 12:30 - 13:00	Sparse differential resultant formulas: between the linear and the nonlinear case S. L. Rueda Wednesday 3, 18:00 - 18:30	Some Optimal Codes as Tanner Codes with BCH Component Codes T. Høholdt, F. Piñero and P. Zeng Friday 5, 9:30 - 10:00	On regular forests given in terms of algebraic geometry, new families of expanding graphs with large girth and new multivariate cryptographic algorithms U. Romanczuk and V. Ustymenko Friday 5, 15:30 - 16:00	Envelope computation in dynamic geometry systems F. Botana and T. Recio Wednesday 3, 17:30 - 18:00	Similarity Detection for Rational Curves J. G. Alcázar, C. Hermoso and G. Muntingh Thursday 4, 12:30 - 13:00
Moving Frames and Noether's Conservation Laws - the General Case T. M. N. Gonçalves and E. L. Mansfield Wednesday 3, 13:00 - 13:30	Seeking recursion operators - an universal hierarchy example in dimension $(2 + 1)$ H. Baran Thursday 4, 10:00 - 10:30	A secret sharing scheme using Gröbner basis H. Kai and M. Yamada Friday 5, 10:00 - 10:30	Some remarks for codes and lattices over imaginary quadratic fields T. Shaska and C. Shor Friday 5, 16:00 - 16:30	Obtaining combinatorial structures associated with low-dimensional Leibniz algebras M. Ceballos, J. Núñez and Á. F. Tenorio Wednesday 3, 18:00 - 18:30	Padovan-like sequences and Bell polynomials N. Gogin and A. Mylläri Thursday 4, 13:00 - 13:30
Interpolation with integral and Stieltjes conditions A. Korporal and G. Regensburger Wednesday 3, 15:00 - 15:30	The algebra of polynomial integro-differential operators and its group of automorphisms V. V. Bavula Thursday 4, 10:30 - 11:00	Error-correcting pairs and arrays from algebraic geometry codes I. Márquez-Corbella and R. Pellikaan Friday 5, 10:30 - 11:00	An Efficient Algorithm for Computing Branch Groebner Systems and Its Applications in Algebraic Cryptanalysis Y. Sun, Z. Huang, D. Lin and D. Wang Friday 5, 17:00 - 17:30		
		Additive multivariable codes over F_4 E. Martínez-Moro, A. P. Nicolás and I. F. Rua Friday 5, 11:30 - 12:00	On some algebraic aspects of data security in cloud computing V. Ustymenko and A. Wroblewska Friday 5, 17:30 - 18:00		

06. Arithmetic of Algebraic Curves Room III	07. Applications and Libraries development in DERIVE and TI-NSPIRE Room I	08. Computer Algebra in Algebraic Statistics (Room III)	09. Computer algebra, quantum computing and quantum information processing Room II	10. Computer algebra in algebraic topology and its applications Room III	11. Symbolic and Numerical Methods: Practical Applications Room II
On the automorphisms groups of Hassett moduli spaces A. Massarenti Thursday 4, 10:00 - 10:30	A Dynamic Unity of Tradition and Technology in Undergraduate Mathematics - a Bulgarian Experience E. A. Varbanoba Saturday 6, 9:30 - 10:00	Goodness-of-fit testing in Ising Models A. Martín del Campo and C. Uhler Friday 5, 11:30 - 12:00	On the inequalities defining the entanglement space of 2-qubits V. P. Gerdt, A. M. Khvedelidze and Y. G. Palii Wednesday 3, 10:00 - 10:30	Spectral Sequences for computing persistent homology of digital images A. Romero, G. Mata, J. Rubio, J. Heras and F. Sergeraert Wednesday 3, 10:00 - 10:30	Kinematical analysis of mechanisms with computer algebra S. Piipponen, T. Arponen and J. Tuomela Friday 5, 10:30 - 11:00 (Room III)
Weierstrass points and dihedral invariants of superelliptic curves C. Shor and T. Shaska Thursday 4, 10:30 - 11:00	A Toolbox with DERIVE: Calculus on Several Variables A. García, F. García, Á. Martín, G. Rodríguez and A. de la Villa Saturday 6, 10:00 - 10:30	Algebraic geometry in causal inference C. Uhler Friday 5, 12:00 - 12:30	Mathematica Package Quantum Circuit for Simulation of Quantum Computation V. P. Gerdt and A. N. Prokopenya Wednesday 3, 10:30 - 11:00	A chain contraction approach to the computation of cubical homology and cohomology P. Pilarczyk and P. Real Wednesday 3, 10:30 - 11:00	Using CUDA for better harnessing of symbolic and numerical methods S. Ortega, J.M. González, M.L. Muñoz, C. Sánchez and T. Morales Saturday 6, 9:30 - 10:00
On the Approximate Parametrization Problem of Algebraic Curves S. L. Rueda, J. R. Sendra and J. Sendra Thursday 4, 11:30 - 12:00	DERIVE and Linear Algebra A. García, F. García, Á. Martín, G. Rodríguez and A. de la Villa Saturday 6, 10:30 - 11:00	Computing real log canonical thresholds in algebraic statistics H. Ahmadinezhad, J. Schicho and C. Uhler Friday 5, 12:30 - 13:00	Simulating quantum computers with Mathematica: QDENSITY et al. B. Julia-Díaz and F. Tabakin Wednesday 3, 11:30 - 12:00	Discrete Morse Theory and Computational Homology P. Dlotko Wednesday 3, 11:30 - 12:00	Resolution of solar cells equivalent electrical model by reverse decomposition S. Merino, F. J. Sánchez, P. Rodríguez and C. Sánchez Saturday 6, 10:00 - 10:30
Radical parametrization of algebraic curves and surfaces D. Sevilla and J. R. Sendra Thursday 4, 12:00 - 12:30	Integration of Piecewise Continuous Functions (Part I, Part II) M. Beaudin, F. Henri and G. Savard Saturday 6, 11:30 - 12:30	Error evaluation for algebraic interpolatory cubature formulae: notable examples C. Fassino and E. Riccomagno Friday 5, 13:00 - 13:30	Functional framework for representing and transforming quantum channels J. A. Miszczak Wednesday 3, 12:00 - 12:30	Some Advanced in G-invariant topology and homology P. Frosini Wednesday 3, 12:00 - 12:30	Statistical Quality Control in the Construction Industry J. A. Vera, S. Merino, J. L. Galán Saturday 6, 10:30 - 11:00
On superelliptic curves and their Jacobians T. Shaska and L. Beshaj Thursday 4, 12:30 - 13:00		Connectivity on two-way tables under certain models R. Yoshida Friday 5, 15:00 - 15:30	Ultimate limits to squeezing of quantum fluctuations A. Orłowski Wednesday 3, 12:30 - 13:00	Computational Homological Algebra for Advanced Topological Analysis of 4D Digital Images P. Real Wednesday 3, 12:30 - 13:00	Optimization and design of bicycles lines R. J. Liñan, S. Merino and J. Martínez Saturday 6, 11:30 - 12:00
Hyperbolic uniformizations through computations on ternary quadratic forms M. Alsina Thursday 4, 13:00 - 13:30	FOPDE.mth: Solving First-Order Partial Differential Equations with DERIVE 6 step by step G. Aguilera, J. L. Galán, M. Á. Galán, Y. Padilla, P. Rodríguez and R. Rodríguez Saturday 6, 12:30 - 13:00	Monomial ideal methods for hierarchical statistical models H. Maruri-Aguilar, E. Sáenz-de-Cabezón and H. P. Wynn Friday 5, 15:30 - 16:00	Geometry and Dynamics of Algorithms on the Quantum Information Space Y. Uwano Wednesday 3, 13:00 - 13:30	A-infinity persistence Kiko Belchi, Aniceto Murillo Mas Wednesday 3, 13:00 - 13:30	Numerical algorithm solves for a new positioning system inside buildings A. B. Pabón, S. Merino and P. Rodríguez Saturday 6, 12:00 - 12:30
From relations in the moduli spaces of curves, to recursions in Gromov-Witten theory N. Pagani Friday 5, 9:30 - 10:00		A Gröbner Bases Method for Complementary Sequences C. Koukouvinos, D. E. Simos and Z. Zafeirakopoulos Friday 5, 16:00 - 16:30		On Higher Dimensional Cocyclic Hadamard Matrices V. Álvarez Solano, J. A. Armario, M. D. Frau and P. Real Wednesday 3, 15:00 - 15:30	CAS software for teaching numerical methods in engineering. Practical applications C. Sánchez, J.M. González, T. Morales, M.L. Muñoz and S. Ortega Saturday 6, 12:30 - 13:00
Symplectic representations for finite group actions on Riemann surfaces A. Behn, A. Rojas and R. Rodríguez Friday 5, 10:00 - 10:30				Boundary and Acyclicity Operators of Primal and Dual Elementary Cell Complexes A. M. Pacheco and P. Real Wednesday 3, 15:30 - 16:00	

Wednesday 3

8:00 - 8:30	REGISTRATION		
08:30 - 09:00	Opening (Room I)		
09:00 - 10:00	Eugenio Roanes Lozano Some ideas about the research in computational mathematics: a perspective from the middle of the life and the academic life Plenary Lecture. Room I		
10:00 - 10:30	03. Algebraic and Algorithmic Aspects of Differential and Integral Operators (Room I)	09. Computer algebra, quantum computing and quantum information processing (Room II)	10. Computer algebra in algebraic topology and its applications (Room III)
	Darboux theory of integrability in the sparse case G. Chèze	On the inequalities defining the entanglement space of 2-qubits V. P. Gerdt, A. M. Khvedelidze and Y. G. Pali	Spectral Sequences for computing persistent homology of digital images A. Romero, G. Mata, J. Rubio, J. Heras and F. Sergeraert
10:30 - 11:00	Isomorphisms and Serre's reduction of multidimensional linear systems T. Cluzeau and A. Quadrat	Mathematica Package Quantum Circuit for Simulation of Quantum Computation V. P. Gerdt and A. N. Prokopenya	A chain contraction approach to the computation of cubical homology and cohomology P. Pilarczyk and P. Real
11:00 - 11:30	COFFEE BREAK		
11:30 - 12:00	Qualitative Study of Polynomial Differential Systems D. Dali and A. Turki	Simulating quantum computers with Mathematica: QDENSITY et al. B. Julia-Diaz and F. Tabakin	Discrete Morse Theory and Computational Homology P. Dlotko
12:00 - 12:30	Periodic and Mean-Periodic Solutions of LODEs with Constant Coefficients I. Dimovski and M. Spiridonova	Functional framework for representing and transforming quantum channels J. A. Miszczak	Some Advanced in G-invariant topology and homology P. Frosini
12:30 - 13:00	Looking for invariant algebraic curves A. Ferragut and A. Gasull	Ultimate limits to squeezing of quantum fluctuations A. Orlowski	Computational Homological Algebra for Advanced Topological Analysis of 4D Digital Images P. Real
13:00 - 13:30	Moving Frames and Noether's Conservation Laws - the General Case T. M. N. Gonçalves and E. L. Mansfield	Geometry and Dynamics of Algorithms on the Quantum Information Space Y. Uwano	A-infinity persistence Kiko Belchí, Aniceto Murillo Mas
13:30 - 15:00	LUNCH		
15:00 - 15:30	Interpolation with integral and Stieltjes conditions A. Korporal and G. Regensburger	05. Nonstandard Applications of Computer Algebra (Room II)	On Higher Dimensional Cocyclic Hadamard Matrices V. Álvarez Solano, J. A. Armario, M. D. Frau and P. Real
		Simulating Car Traffic with Smart Signals using a CAS J. L. Galán, G. Aguilera, J. C. Campos and P. Rodríguez	
15:30 - 16:00	On Completely Integrable Pfaffian Systems with Normal Crossings S. Maddah	Population-based anamorphosis maps for railway radial networks E. Roanes-Lozano, A. García-Álvarez, J. L. Galán-García and L. Mesa	Boundary and Acyclicity Operators of Primal and Dual Elementary Cell Complexes A. M. Pacheco and P. Real
16:00 - 16:30	Linear Boundary Problems for Partial Differential Equations: Algebraic Setup and First Steps for Constant Coefficients N. Phisanbut and M. Rosenkranz	An algebraic approach to geometric proof using a Computer Algebra System M. Xue	00. General Session on Computer Algebra (Room III)
			A New Criterion for the Existence of Real Zeros of Polynomial Systems. J. S. Cheng
16:30 - 17:00	COFFEE BREAK		
17:00 - 17:30	On the arithmetic of d'Alembertian functions C. Raab	Designing Hamiltonian Cycles F. de Arriba, E. Corbacho and R. Vidal	Planar arrangements and singular algebraic surfaces. J. García Escudero
17:30 - 18:00	Applying Thomas decomposition and algebraic analysis to certain nonlinear PDE systems D. Robertz	Envelope computation in dynamic geometry systems F. Botana and T. Recio	Computations of Gröbner-Shirshov Basis and Reduced Words for Affine Weyl Group $\tilde{A}_n$ using Mathematica. E. Yilmaz, Cenap Özel and Ugur Ustaoglu
18:00 - 18:30	Sparse differential resultant formulas: between the linear and the nonlinear case S. L. Rueda	Obtaining combinatorial structures associated with low-dimensional Leibniz algebras M. Ceballos, J. Núñez and Á. F. Tenorio	
18:30 - 19:00			
19:00	Excursion Malaga city centre and Dinner in Restaurant "El Pimpi"		

Wednesday 3

Thursday 4

09:00 - 10:00	Ilias Kotsireas 10 years of sequence autocorrelation Plenary Lecture (Room I)		
10:00 - 10:30	03. Algebraic and Algorithmic Aspects of Differential and Integral Operators (Room I) Seeking recursion operators - an universal hierarchy example in dimension $(2 + 1)$ H. Baran	05. Nonstandard Applications of Computer Algebra (Room II) A hybrid expert system for classic car recognition and originality evaluation E. Roanes-Lozano and J. Bonilla	06. Arithmetic of Algebraic Curves (Room III) On the automorphisms groups of Hassett moduli spaces A. Massarenti
	The algebra of polynomial integro-differential operators and its group of automorphisms V. V. Bavula	Designing rotating schedules by using Gröbner bases R. Falcón, D. Canca and E. Barrena	Weierstrass points and dihedral invariants of superelliptic curves C. Shor and T. Shaska
11:00 - 11:30	COFFEE BREAK		
11:30 - 12:00	04. Computer Algebra in Coding Theory and Cryptography (Room I) A characterization of cyclic codes whose minimum distance equals their maximum BCH bound J. J. Bernal, D. H. Bueno and J. J. Simon	Flexibility of Structures via Computer Algebra R. H. Lewis and E. Coutsiás	On the Approximate Parametrization Problem of Algebraic Curves S. L. Rueda, J. R. Sendra and J. Sendra
	Gröbner Bases and Linear Codes over Prime Fields N. Dück and K. H. Zimmermann		
12:00 - 12:30	Gröbner Bases and Linear Codes over Prime Fields N. Dück and K. H. Zimmermann	Modeling reliability in propositions using computer algebra techniques A. Hernando	Radical parametrization of algebraic curves and surfaces D. Sevilla and J. R. Sendra
12:30 - 13:00	A Class of Binary Sequences with Large Linear Complexity A. Fúster Sabater	Similarity Detection for Rational Curves J. G. Alcázar, C. Hermoso and G. Muntingh	On superelliptic curves and their Jacobians T. Shaska and L. Beshaj
13:00 - 13:30	Further Improvements on the Feng-Rao Bound for Dual Codes O. Geil and S. Martin	Padovan-like sequences and Bell polynomials N. Gogin and A. Mylläri	Hyperbolic uniformizations through computations on ternary quadratic forms M. Alsina
13:30 - 15:00	LUNCH		
15:00	Excursion to the Alhambra of Granada with dinner		

Thursday 4

Friday 5

	04. Computer Algebra in Coding Theory and Cryptography (Room I)	01. Computer Algebra in Education (Room II)	06. Arithmetic of Algebraic Curves (Room III)
9:30 - 10:00	Some Optimal Codes as Tanner Codes with BCH Component Codes T. Høholdt, F. Piñero and P. Zeng	Computer Algebra - the engine of transition to activity-based approach in mathematics education E. Varbanova and E. Shoikova	From relations in the moduli spaces of curves, to recursions in Gromov-Witten theory N. Pagani
10:00 - 10:30	A secret sharing scheme using Gröbner basis H. Kai and M. Yamada	Assessing Mathematical Content in a Technology Environment Discussion Panel A. Homero Flores	Symplectic representations for finite group actions on Riemann surfaces A. Behn, A. Rojas and R. Rodríguez
10:30 - 11:00	Error-correcting pairs and arrays from algebraic geometry codes I. Márquez-Corbella and R. Pellikaan		11. Symbolic and Numerical Methods: Practical Applications (Room III)
			Kinematical analysis of mechanisms with computer algebra S. Piipponen, T. Arponen and J. Tuomela
11:00 - 11:30	COFFEE BREAK		
11:30 - 12:00	Additive multivariable codes over F4 E. Martínez-Moro, A. P. Nicolás and I. F. Rua	CAS: A Tool for Improving Autonomous Work A. García, F. García, Á. Martín, G. Rodríguez and A. de la Villa	08. Computer Algebra in Algebraic Statistics (Room III)
			Goodness-of-fit testing in Ising Models A. Martín del Campo and C. Uhler
12:00 - 12:30	On Generalized Lee Weigth Codes over Dihedral Groups E. Martínez-Moro, A. P. Nicolás and E. Suárez-Canedo	Investigating Magic Squares in a Linear Algebra Course K. Schmidt	Algebraic geometry in causal inference C. Uhler
12:30 - 13:00	Decoding of codes for applications to steganography C. Munuera and W. Olaya León	Polynomial Systems Solving with Nspire CAS (Part I, Part II) M. Beaudin, G. Picard and G. Savard	Computing real log canonical thresholds in algebraic statistics H. Ahmadinezhad, J. Schicho and C. Uhler
13:00 - 13:30	On LDPC codes corresponding to new families of regular expanding graphs of large girth M. Polak and V. Ustimenko		Error evaluation for algebraic interpolatory cubature formulae: notable examples C. Fassino and E. Riccomagno
13:30 - 15:00	LUNCH		
15:00 - 15:30	Representation, constructions and minimum distance computation of binary nonlinear codes J. Pujol, M. Villanueva and F. Zeng	In praise of rectangular systems D. Jeffrey	Connectivity on two-way tables under certain models R. Yoshida
15:30 - 16:00	On regular forests given in terms of algebraic geometry, new families of expanding graphs with large girth and new multivariate cryptographical algorithms U. Romanczuk and V. Ustymenko	Omega - a free online Computer Algebra System Explorer M. Xue	Monomial ideal methods for hierarchical statistical models H. Maruri-Aguilar, E. Sáenz-de-Cabezón and H. P. Wynn
16:00 - 16:30	Some remarks for codes and lattices over imaginary quadratic fields T. Shaska and C. Shor	Some maths problems for the average citizen E. Roanes-Lozano and J. Cabezas-Corchero	A Gröbner Bases Method for Complementary Sequences C. Koukouvinos, D. E. Simos and Z. Zafeirakopoulos
16:30 - 17:00	COFFEE BREAK		
17:00 - 17:30	An Efficient Algorithm for Computing Branch Groebner Systems and Its Applications in Algebraic Cryptanalysis Y. Sun, Z. Huang, D. Lin and D. Wang	Using Computer Algebra in Mathematics for Engineers T. Westermann	02. Computer Algebra for Dynamical Systems and Celestial Mechanics (Room III)
			The Study of Isochronicity and Critical Period Bifurcations on Center Manifolds of 3-dim Polynomials Systems Using Computer Algebra M. Mencinger and B. Fercec
17:30 - 18:00	On some algebraic aspects of data security in cloud computing V. Ustymenko and A. Wroblewska	Global application of CAS tools for teaching in Computer Engineering degrees S. Cárdenas, I. Fortes, I. Pérez de Guzmán, S. Sánchez and A. Valverde	On using computer algebra systems for analysis of rigid body dynamics L. Burlakova and V. Irtegov
18:00 - 19:30	ACA-WG Meeting		
19:30 - 21:00			
21:00	Gala Dinner		

Friday 5

Saturday 6

9:30 - 10:00	07. Applications and Libraries development in DERIVE and TINSPIRE (Room I)	11. Symbolic and Numerical Methods: Practical Applications (Room II)	02. Computer Algebra for Dynamical Systems and Celestial Mechanics (Room III)
	A Dynamic Unity of Tradition and Technology in Undergraduate Mathematics - a Bulgarian Experience E. A. Varbanova	Using CUDA for better harnessing of symbolic and numerical methods S. Ortega, J.M. González, M.L. Muñoz, C. Sánchez and T. Morales	Formal Integral and Caustics in Henon-Heiles model T. Myllari and A. Myllari
	A Toolbox with DERIVE: Calculus on Several Variables A. García, F. García, Á. Martín, G. Rodríguez and A. de la Villa	Resolution of solar cells equivalent electrical model by reverse decomposition S. Merino, F. J. Sánchez, P. Rodríguez and C. Sánchez	On necessary conditions of integrability of degenerated planar ODE systems in the parameter space A. Bruno and V. Edneral
10:00 - 10:30	DERIVE and Linear Algebra A. García, F. García, Á. Martín, G. Rodríguez and A. de la Villa	Statistical Quality Control in the Construction Industry J. A. Vera, S. Merino, J. L. Galán	Tree structures in Poisson series processors J. F. Navarro
10:30 - 11:00	COFFEE BREAK		
11:00 - 11:30	Integration of Piecewise Continuous Functions (Part I, Part II) M. Beaudin, F. Henri and G. Savard	Optimization and design of bicycles lines R. J. Liñan, S. Merino and J. Martínez	Orbital Reversibility of Dynamical Systems A. Algaba, I. Checa, C. García and E. Gamero
11:30 - 12:00		Numerical algorithm solves for a new positioning system inside buildings A. B. Pabón, S. Merino and P. Rodríguez	Normal Forms of Singular Plane Quartics T. Takahashi
12:00 - 12:30	FOPDE.mth: Solving First-Order Partial Differential Equations with DERIVE 6 step by step G. Aguilera, J. L. Galán, M. Á. Galán, Y. Padilla, P. Rodríguez and R. Rodríguez	CAS software for teaching numerical methods in engineering. Practical applications C. Sánchez, J.M. González, T. Morales, M.L. Muñoz and S. Ortega	Random walks on the three dimensional Young tree N. Vassiliev
12:30 - 13:00	Closing Ceremony		
13:00 - 13:30			

Saturday 6